

Wi-fi on wheels

The 2010 Jeep Grande Cherokee features a Uconnect Web, which is a WiFi hotspot allowing you to surf from inside the car

24-in-1 card reader

Lexar's has released a 24-in-1 card reader that can hook up to the Apple iPad through its camera connection kit allowing importing pictures to iPad

3D in your hands

Nintendo's plans were to make 3D a reality for the mass market without making people invest thousands of dollars, the Nintendo 3DS is their solution. An upgrade of the Nintendo DS, the immensely popular handheld gaming device, the Nintendo 3DS can display 3D images and video without the use of special glasses. Lack of glasses and the affordable price tag of the 3DS is also a huge push for the mass popularisation of 3D. 3DS was Nintendo's trump card at E3 2010.

Nintendo 3DS looks just like a DSi, with two screens, a 3.5-inch 3D capable widescreen on top, with a smaller touch-screen at the bottom. Apart from adding 3D, Nintendo has

is no gimmick, and the screen actually works. 'Parallax barrier' is the technology behind the glasses-free or auto-stereoscopic screen of the Nintendo 3DS (*Editor's Note: that the 3DS uses parallax barrier technique is an educated guess as this has not been officially revealed to be the case*). This screen has a coating of what is called the 'parallax barrier aperture array', which is a set of vertical black bars provides a separate 'window' for each eye to view alternate pixels, which is like doing the job of googles at the display level. However, the viewing angle is limited in an auto-stereoscopic screen. The parallax barrier coating is actually like another screen and the spacing between the 'black bars' can be changed to



Nintendo3DS

Powerful new GPU

As the 3D effect requires rendering of the scene twice, from two different angles, the 3DS is very graphic-intensive. The difficult job of being the Graphics Processing Unit (GPU) of this new console is done by Digital Media Professional's (DMP) Pica200 GPU. The graphics core achieves high performance with low power consumption. PICA200 is an OpenGL ES1.1 chip with DMP's proprietary extension graphics API called Maestro, the combination making it an equivalent of an OpenGL ES2.0 chip. This API enables hardware-based graphics rendering, including the T & L (transform and lighting), texture processing and full screen anti-aliasing, among a host of other post-processing effects. The other noticeable factor is the ultra-high polygon rendering rate, adding up to 15.3 million polygons per second at a native clock speed of 200MHz.

Other new features include a camera with two lenses 3D photos. The screen can support playback of 3D movies, and there might be possible

content tie-ups with Disney, WB, and DreamWorks. The Nintendo 3DS features wireless functionality, where it can communicate with another 3DS, even when it's in sleep mode. The device will also retain compatibility with its predecessor, the DSi. It can also access the internet through WiFi, and get downloadable content. There's apparently no monthly fee required for this.

The titles announced at launch for the 3DS include :

- Kid Icarus Uprising (Out in the "summer")
- DJ Hero 3D
- Kingdom Hearts 3D
- Saints Row 3D
- Madden
- FIFA
- Resident Evil Revelations
- Assassin's Creed
- Metal Gear Solid
- Final Fantasy
- Codename: Chocoboco Racing 3D
- Super Street Fighter IV

Neither the release date, nor the price were unveiled at the time of printing, but the rumours are of a December release with launch price between \$249.99 and \$299.99. 



A demo frame rendered using Pica200

also made some other hardware changes to the DS to up its graphics capabilities, and give the player better control with the new accelerometer and gyroscope, and analog stick.

The wonder-screen

We have not had a hands-on with the 3DS, but those who have had a look say that the 3D

change the apparent depth of the 3D images. This is done by the '3D depth slider' provided on the side of the console. This slider also allows a person to switch to 2D whenever 3D gets too heavy on the eyes, as it often can. The 3D screen has a resolution of 800x240, which effectively becomes 400x240 when viewing 3D images.